E-COMMERCE SALES ANALYSIS: CUSTOMER BEHAVIOUR, PRODUCT PERFORMANCE AND SALES TRENDS

A Python-Based Data Analysis Project

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1. Abstract

E-Commerce Sales Analysis: Customer Behaviour, Product Performance and Sales Trends focuses on analysing e-commerce transaction data to understand customer purchasing patterns, evaluate product performance and identify sales trends.

The project uses a dataset obtained from Kaggle containing 49,297 records and 17 columns. Python was used for data cleaning, Exploratory Data Analysis (EDA), statistical analysis and data visualisation. The preprocessing stage involved handling missing values, removing duplicates, correcting data types, standardising categorical values and identifying potential outliers using the Interquartile Range (IQR) method.

The analysis examined sales by product category, customer segment and city, along with monthly sales trends, payment methods, order statuses and relationships between numerical variables. Visualisations were created using Matplotlib, Seaborn and Plotly to present the findings clearly.

The results showed total recorded sales of 3,445,750, with Electronics contributing the highest sales among product categories. Tehran recorded the highest sales among the cities analysed, and Gateway was the most commonly used payment method. Correlation analysis also revealed a strong positive relationship between unit price and sales, and a moderate positive relationship between quantity and sales.

Overall, the project demonstrates how data analysis and visualisation can reveal useful patterns in e-commerce transactions and support informed business decision-making.

2. Introduction

E-commerce has become an important part of modern business, generating large amounts of data through online transactions, customer purchases and product sales. Analysing this data helps businesses understand customer behaviour, evaluate product performance and identify changes in sales trends.
Understanding these patterns is important for businesses to recognise customer preferences, identify popular products, compare market performance and make informed decisions. Sales data can also reveal differences in purchasing behaviour across customer segments and geographical locations.
The project, E-Commerce Sales Analysis: Customer Behaviour, Product Performance and Sales Trends, focuses on exploring these aspects of e-commerce transactions. It aims to identify meaningful patterns in customer purchasing behaviour, product sales and sales trends to develop a better understanding of business performance and potential areas for improvement.

3. Problem Statement

E-commerce businesses generate large volumes of transaction data, but understanding this data is essential for identifying purchasing patterns, evaluating product performance and monitoring sales trends. Without a clear analysis of this information, it can be difficult to determine which product categories contribute the most to sales, how purchasing behaviour differs across customer segments and cities, and which payment methods are most commonly used.
The project addresses the need to analyse e-commerce transaction data systematically to identify meaningful patterns, relationships and variations in sales performance. It also seeks to examine how factors such as quantity and unit price relate to recorded sales, helping to develop a clearer understanding of transaction behaviour.
Therefore, the main problem addressed by this project is how to transform raw e-commerce transaction data into meaningful insights that can support better understanding of customer behaviour, product performance and sales trends.

4. Project Objectives

- To analyse overall sales performance by examining total recorded sales and identifying patterns across the dataset.
- To evaluate product and category performance by comparing sales values and quantities sold across different products and product categories.
- To understand customer behaviour by examining customer age distribution, customer segments and purchasing patterns.
- To analyse geographical sales patterns by comparing recorded sales across different cities.
- To examine payment methods and order statuses to understand payment preferences and the distribution of completed, cancelled and returned orders.
- To identify sales trends over time by analysing monthly sales patterns and comparing trends across different years.
- To investigate pricing and quantity relationships by examining how unit price, quantity and discount relate to recorded sales.

- To identify relationships between numerical variables using statistical analysis and correlation measures.
- To present key findings through data visualisations and develop insights that may support informed business decision-making.

5. Dataset Description

5.1 Dataset Source

The dataset used in this project, *E-Commerce Sales Data Analysis & EDA*, was obtained from Kaggle. It contains e-commerce transaction records covering customer information, product details, sales values, payment methods and order statuses.
The dataset was selected because it provides a combination of numerical, categorical and date-related variables, making it suitable for analysing customer behaviour, product performance and sales trends.

5.2 Dataset Dimensions

The original dataset contained 49,297 rows and 17 columns. Each row represents a transaction record, while each column provides a specific piece of information about the transaction, customer, product or order.
The dataset includes numerical variables, such as Quantity, Age, UnitPrice, OrderValue and Sales; categorical variables, such as Category, PaymentMethod, Status and CustomerSegment; and date-related variables, such as OrderDate and SignupDate.
These different types of information allow the dataset to be examined from multiple perspectives, including sales performance, customer purchasing patterns, geographical differences and changes in sales over time.

5.3 Description of Important Columns

The dataset contains the following important columns used to understand and analyse e-commerce transactions.

Table 1. Description of Important Dataset Columns

Column Name	Description
OrderID	Identifier associated with an order or transaction record.
CustomerID	Identifier used to distinguish customers.
OrderDate	Date on which the order was recorded.
ProductID	Identifier associated with a product.
Quantity	Number of units recorded in a transaction.
Discount	Discount value recorded for the transaction.
PaymentMethod	Payment method recorded for the transaction.
Status	Order status, including Completed, Cancelled and Returned.
Age	Age of the customer.
City	City recorded in the dataset.
SignupDate	Date on which the customer registered.

CustomerSegment	Customer classification, such as New, Regular or VIP.
ProductName	Name of the product.
Category	Product category, such as Electronics, Accessories or Gaming.
UnitPrice	Price per unit of the product.
Sales	Recorded sales value associated with the transaction.
OrderValue	Order value recorded in the dataset. This column was later found to contain the same values as Sales and was removed during preprocessing.

6. Tools and Technologies Used

6.1 Google Colab

Google Colab is a cloud-based platform that allows users to write and execute Python code in an interactive notebook. It was used as the working environment for this project, allowing data loading, cleaning, exploratory data analysis and visualisation to be performed in one place.

6.2 Python

Python is a programming language widely used for data analysis and automation. In this project, Python was used to process the e-commerce dataset, perform calculations, examine patterns and generate visualisations.

6.3 Pandas

Pandas is a Python library used for data manipulation and analysis. It provides DataFrames for organising tabular data and functions for handling missing values, removing duplicates, changing data types, filtering records, grouping data and calculating summary statistics. It was a key library for cleaning and analysing the dataset.

6.4 NumPy

NumPy is a Python library that supports numerical calculations and operations on arrays. It was used to support numerical analysis and calculations involved in examining the dataset.

6.5 Matplotlib

Matplotlib is a Python library used to create static charts and graphs. It was used to present findings through visualisations, including bar charts and other plots, making comparisons across product categories, cities and customer segments easier to understand.

6.6 Seaborn

Seaborn is a statistical data visualisation library built on Matplotlib. It provides functions for creating informative statistical graphics. In this project, it was used to create visualisations such as the customer age distribution, sales box plot, correlation heatmap, grouped count plot and scatter plot.

6.7 Plotly

Plotly is a data visualisation library that supports interactive charts. It was used to create visualisations such as the top 10 products by sales, monthly sales trends and payment method distribution. Its interactive features allow users to explore chart details more easily.
These tools and libraries were used together to prepare the dataset, conduct exploratory data analysis and present the findings in a clear and understandable way.

7. Data Loading and Initial Inspection

7.1 Data Loading

The e-commerce dataset was obtained from Kaggle and imported into Google Colab using the Pandas library. The pd.read_csv() function was used to load the CSV file into a DataFrame named df, which served as the primary data structure for subsequent data inspection, preprocessing and analysis.
Before examining the dataset, the project title, dataset source and other relevant details were documented. The dataset's features were also organised into logical categories, including transaction, product, customer and order information, to establish an overview of the variables available for analysis.

7.2 Initial Data Inspection

Initial data inspection was performed to understand the dataset's structure, dimensions, data types and overall quality before applying any cleaning procedures. Several Pandas methods and attributes were used for this purpose.

- df and df.head() – Dataset Preview: The DataFrame was displayed, and the head() method was used to examine the first five records. This provided an initial overview of the column names, values and structure of the dataset.
- df.tail() – Final Records: The tail() method displayed the last five records, allowing the end of the dataset to be examined and helping verify that the data had been loaded through to its final rows.
- df.info() – Structural Assessment: The info() method provided a concise summary of the DataFrame, including column names, data types and non-null counts. This helped identify potential data type inconsistencies and columns requiring further investigation for missing values.
- df.shape – Dataset Dimensions: The shape attribute returned the number of rows and columns in the dataset. The original dataset contained 49,297 rows and 17 columns, establishing the baseline dimensions before preprocessing.
- df.dtypes – Data Type Assessment: The dtypes attribute displayed the data type of each column. This helped determine whether variables were stored in appropriate formats and identify columns that might require conversion, particularly date-related fields.

- df.describe() – Descriptive Statistics: The describe() method generated summary statistics for numerical variables, including the count, mean, standard deviation, minimum, maximum and quartile values. These statistics provided an initial understanding of the distributions, ranges and variability of numerical features.
- df.isnull().sum() – Missing Value Assessment: This expression calculated the number of missing values in each column, helping identify variables that required attention during data cleaning.
- df.duplicated().sum() – Duplicate Record Assessment: This expression counted duplicate rows in the DataFrame, providing an initial indication of potential record duplication and the need for further investigation.

7.3 Summary of Initial Inspection

The initial inspection established the dataset's original dimensions and provided an overview of its structure, data types and numerical characteristics. Missing-value and duplicate checks were also conducted to assess data completeness and identify potential quality issues.
The results of these preliminary checks informed the subsequent data cleaning and preprocessing stage, during which identified issues were investigated and addressed using appropriate methods.

8. Data Cleaning and Preprocessing

Data cleaning and preprocessing were performed to improve the quality, consistency and suitability of the e-commerce dataset for exploratory data analysis. The process involved handling missing values, removing duplicate records, standardising categorical variables, converting data types, investigating invalid transaction values, removing redundant information, identifying potential outliers and creating additional analytical features.

8.1 Handling Missing Values

The initial inspection identified **250 missing values** across four columns. Appropriate imputation methods were selected according to the nature of each variable to preserve the available records while addressing incomplete information.

Table 2. Missing-Value Treatment

Column	Missing values identified	Treatment Method
Age	150	Replaced with median
Discount	50	Replaced with median
PaymentMethod	20	Replaced with the mode
City	30	Replaced with Unknown
Total	250	All missing values addressed

The median was used for Age and Discount because it is less affected by extreme values than the mean. This was appropriate for numerical variables where unusually high or low observations could influence the average. The mode was used for PaymentMethod because it represents the most frequently occurring category. For City, missing entries were labelled Unknown rather than assigned an assumed location, preserving the records without introducing unsupported geographical information.
Following imputation, the missing-value check returned zero missing values across the dataset.

8.2 Removing Duplicate Records

Exact duplicate records were investigated using Pandas' duplicated() method. The initial check identified **75 duplicate rows**, which were targeted for removal using drop_duplicates(). A subsequent duplicate check returned zero, indicating that no duplicate rows remained in the checked DataFrame.
Removing exact duplicates helps prevent repeated records from being counted more than once and reduces the risk of distorting summary statistics and visualisations.

8.3 Standardising Categorical Values

Categorical variables were examined for inconsistencies caused by differences in capitalisation and leading or trailing spaces. These inconsistencies can cause equivalent categories to be treated as separate groups during analysis.
The PaymentMethod, CustomerSegment and Category columns were standardised by trimming unnecessary spaces and correcting text capitalisation. City names were also processed to remove leading and trailing spaces.

Table 3. Categorical Standardisation

Column	Inconsistencies identified	Standardisation applied
PaymentMethod	8 non-standard label variants across four payment methods	Corrected capitalisation and standardised labels
CustomerSegment	6 non-standard label variants across three customer segments	Standardised capitalisation and spacing
Category	4 non-standard label variants across the six product categories	Removed surrounding spaces and standardised capitalisation
City	Potentially inconsistent surrounding spaces	Removed leading and trailing spaces

After standardisation, PaymentMethod contained four categories: Gateway, CardToCard, Wallet and Cash. CustomerSegment contained three categories: Regular, New and VIP. The product category values were consolidated into six categories: Accessories, Electronics, Stationery, Home Office, Wearables and Gaming.
This step ensured that equivalent labels were grouped consistently when comparing sales, quantities and transaction patterns.

8.4 Correcting Data Types

The original data types were examined to determine whether the columns were stored in suitable formats for analysis. The OrderDate and SignupDate columns were initially stored as object data types.
Both columns were converted to datetime64[ns] using Pandas' to_datetime() function.
This conversion enabled date-based operations, including extracting years and months, grouping transactions chronologically and examining changes in sales over time. The data types were checked again after conversion to confirm that the date columns had been updated successfully.

8.5 Identifying and Handling Invalid Transaction Values

The Quantity, Sales and OrderValue columns were examined for negative values, and the corresponding Status values were investigated to determine whether the records represented potentially invalid completed transactions or cancellation-related adjustments.

The inspection identified 19 records with negative quantities: 18 were associated with the Completed status, while one was associated with the Cancelled status.

The 18 completed records were removed because their negative quantities and negative sales values were inconsistent with the expected representation of completed sales transactions in this analysis. After removal, the verification check returned zero completed transactions with negative quantities.

The remaining cancelled record was retained because negative values may represent a cancellation or reversal rather than an invalid completed sale.

This approach avoided treating all negative values as errors without considering their associated order status.

8.6 Removing Redundant Columns

The Sales and OrderValue columns were compared to determine whether they contained distinct information. The comparison confirmed that their values were identical across the dataset.

Consequently, OrderValue was removed because it duplicated information already available in Sales. Retaining both columns would have added redundancy without providing additional analytical value.

After removing OrderValue, the dataset contained 49,204 rows and 16 columns, according to the displayed DataFrame dimensions.

8.7 Identifying Potential Outliers Using the IQR Method

Potential outliers were investigated using the Interquartile Range (IQR) method. The IQR measures the spread of the middle 50% of observations and can be used to flag values that fall unusually far from the central distribution.

The IQR was calculated as follows:

IQR = Q3 − Q1

The lower and upper boundaries were then calculated using:

- Lower boundary = Q1 − 1.5 × IQR
- Upper boundary = Q3 + 1.5 × IQR

Values outside these boundaries were flagged as potential outliers.

Table 4. Potential Outliers Identified

Column	Potential outliers
Sales	3879
UnitPrice	1691
Quantity	4901

For the Sales column, the calculated lower boundary was −83.5 and the upper boundary was 184.5.

The flagged observations were retained rather than automatically removed. This decision was made because unusually large or small values may represent legitimate transactions, and removing them

without further evidence could distort the analysis. The IQR method was therefore used as a screening technique to identify values requiring attention, rather than as proof that those observations were erroneous.

8.8 Creating Additional Features

Four derived features were created to support time-based analysis and customer segmentation.

Table 5. Derived Features

New column	Derived from	Purpose
OrderYear	OrderDate	Enables annual sales comparisons
OrderMonth	OrderDate	Provides a numerical month value for grouping
OrderMonthName	OrderDate	Makes monthly results easier to interpret in visualisations
AgeGroup	Age	Groups customers into age ranges for demographic analysis

The OrderYear, OrderMonth and OrderMonthName columns were created using the datetime properties of OrderDate. The AgeGroup column grouped customers into five ranges: 18–25, 26–35, 36–45, 46–55 and 56–65.
These features provided additional ways to examine sales patterns over time and compare customer age groups without altering the underlying meaning of the original variables.

8.9 Validation and Final Dataset

The final dataset was checked to confirm its dimensions following the cleaning and preprocessing operations. The changes in dataset size at each major stage are summarised in Table 6.

Table 6. Dataset Dimensions Throughout Preprocessing

Processing stage	Rows	Columns
Original dataset	49297	17
After duplicate removal	49222	17
After removing invalid transactions	49204	17
After removing OrderValue	49204	16
Final dataset after feature creation	49204	20

The final dataset contained **49,204 rows and 20 columns**, providing the data structure required for the subsequent exploratory data analysis. The reduction in rows resulted from the removal of duplicate records and invalid completed transactions, while the change in column count reflected the removal of redundant information and the creation of four additional analytical features.

9. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted on the cleaned e-commerce dataset to examine overall sales performance, product demand, customer characteristics, geographical differences, transaction behaviour and relationships between numerical variables. The analysis combined descriptive statistics, grouped calculations, comparisons across categories and time periods, and correlation analysis.
The following subsections present the numerical results and explain the patterns identified in the dataset.

9.1 Overall Performance Analysis

The overall performance analysis established a baseline understanding of the transaction records by examining total recorded sales, transaction activity, customer coverage and quantity sold.

Table 7. Overall E-Commerce Performance

Metric	Result
Total recorded sales	3,445,750
Transaction records	49,204
Unique order IDs	49,204
Unique customers	9,820
Total quantity recorded	92,658

The dataset contained 49,204 transaction records associated with 9,820 unique customers. The total quantity recorded was 92,658 units, while the sum of recorded sales was 3,445,750.
The number of unique order IDs matched the number of transaction records in the cleaned dataset. This indicates that each remaining record had a distinct order ID, although it does not independently establish whether each record represents a complete customer purchasing journey.
These figures provide a baseline for examining differences in sales performance across products, customers, cities and time periods.

9.2 Product and Category Analysis

Product and category analysis examined how recorded sales and quantities varied across product categories and individual products. This allowed comparisons between sales value and sales volume, helping to identify which products and categories contributed most to the recorded sales total.

9.2.1 Sales by Product Category

Table 8. Total Recorded Sales by Product Category

Product category	Recorded sales
Electronics	1,751,491
Accessories	553,158
Home Office	446,065
Wearables	429,497
Gaming	198,302
Stationery	67,237
Total	**3,445,750**

Electronics generated the highest recorded sales, totalling 1,751,491, which represented approximately 50.8% of the overall recorded sales. Accessories ranked second at 553,158, followed by Home Office at 446,065.
Stationery recorded the lowest sales value at 67,237. The difference between Electronics and the other categories indicates that recorded sales were concentrated in a relatively small number of product categories.
This finding suggests that Electronics was the leading category by sales value in the dataset. However, sales value alone does not establish profitability because product costs and other expenses were not included in the analysis.

9.2.2 Top Products by Recorded Sales

Table 9. Top 10 Products by Recorded Sales

Rank	Product	Recorded sales
1	Headphones	333,531
2	Office Chair	329,292
3	Tablet	292,344
4	Smart Watch	259,590
5	Microphone	230,309
6	Mechanical Keyboard	229,676
7	Bluetooth Speaker	213,427
8	Gaming Controller	198,302
9	Power Bank	180,004
10	Fitness Band	169,907

Headphones recorded the highest sales among individual products at 333,531, followed closely by Office Chair at 329,292. Tablet ranked third at 292,344.
The results show that the leading products included electronics, computer accessories and wearable devices. However, a product's recorded sales value should be distinguished from its sales volume, since a product can generate a high sales value without having the highest number of units sold.

9.2.3 Quantity Sold and Average Unit Price

The quantity analysis indicated that Accessories had the highest sales volume, followed by Electronics. Stationery ranked third, while Home Office and Wearables had similar quantities sold. Gaming recorded the lowest quantity.
Average unit prices also varied across categories.

Table 10. Average Unit Price by Product Category

Product category	Average unit price
Wearables	81.73
Home Office	80.97
Gaming	68.00
Electronics	57.13
Accessories	17.35
Stationery	7.00

Wearables and Home Office had the highest average unit prices, while Stationery had the lowest. Accessories generated the second-highest recorded sales value and the highest quantity sold, despite its comparatively low average unit price.
Together, these findings demonstrate that sales value, sales volume and average unit price provide different perspectives on product performance. The category with the highest sales value does not necessarily have the highest quantity sold or the highest average unit price.

9.3 Customer Analysis

Customer analysis examined the age distribution and customer segments represented in the dataset. The purpose was to understand the characteristics of the recorded customer base and compare sales contributions across different groups.

9.3.1 Customer Age Distribution

The age analysis produced the following summary statistics.

Table 11. Descriptive Statistics for Customer Age

Statistic	Result
Mean age	41.28 years
Median age	41 years
Minimum age	18 years
Maximum age	65 years

The mean customer age was 41.28 years, while the median was 41 years. The close relationship between these values indicates that the centre of the age distribution was around 41 years.
Customer ages ranged from 18 to 65 years. To support more detailed demographic analysis, customers were grouped into five age ranges: 18–25, 26–35, 36–45, 46–55 and 56–65.
These groups provided a consistent basis for comparing age-related patterns without examining every individual age separately. Age distribution alone, however, does not indicate which age group contributed the highest sales.

9.3.2 Sales by Customer Segment

Table 12. Total Recorded Sales by Customer Segment

Customer segment	Recorded sales
Regular	1,915,222
New	1,196,503
VIP	334,025
Total	**3,445,750**

Regular customers contributed the highest recorded sales, totalling 1,915,222, followed by New customers at 1,196,503. VIP customers contributed 334,025.
Regular customers accounted for approximately 55.6% of total recorded sales, while New customers contributed approximately 34.7%. VIP customers represented approximately 9.7%.
These figures indicate that the Regular segment accounted for the largest share of recorded sales. However, total sales alone do not establish that individual Regular customers spent more than customers in other segments. The number of customers in each segment and average sales per customer must also be considered when comparing purchasing behaviour.

9.4 Geographical Analysis

Geographical analysis compared recorded sales across cities to identify differences in sales distribution and customer activity.

9.4.1 Recorded Sales by City

Table 13. Recorded Sales by City

City	Recorded sales
Tehran	963,320
Mashhad	438,089
Karaj	350,030
Isfahan	334,459
Shiraz	310,926
Tabriz	292,153
Ahvaz	211,181
Rasht	205,633
Kerman	175,144
Qom	162,760

Tehran recorded the highest sales at 963,320, substantially exceeding the totals recorded in the other cities. Mashhad ranked second at 438,089, followed by Karaj at 350,030.
Qom recorded the lowest sales among the listed cities at 162,760. Tehran's recorded sales represented approximately 28.0% of the overall total.
The results indicate that recorded sales were geographically concentrated, with Tehran contributing the largest share. However, high total sales may reflect a larger customer base rather than higher spending by individual customers. Comparing unique customer counts and average sales per customer provides additional context for interpreting these differences.

9.5 Payment Method and Order Status Analysis

Payment methods and order statuses were examined to understand transaction frequency and the distribution of recorded order outcomes.

9.5.1 Payment Method Distribution

Table 14. Transaction Records by Payment Method

Payment method	Records	Approximate share
Gateway	23,891	48.6%
CardToCard	11,738	23.9%
Wallet	8,655	17.6%
Cash	4,920	10.0%
Total	**49,204**	**100.0%**

Gateway was the most frequently recorded payment method, accounting for approximately 48.6% of transaction records. CardToCard represented approximately 23.9%, followed by Wallet at 17.6% and Cash at 10.0%.

The results indicate that Gateway was the dominant payment method in the dataset, while Cash was the least frequently recorded. These percentages describe transaction frequency rather than the monetary value processed through each payment method.

9.5.2 Order Status Distribution

Table 15. Transaction Records by Order Status

Order status	Records	Approximate share
Completed	45,258	92.0%
Cancelled	2,438	5.0%
Returned	1,508	3.1%
Total	**49,204**	**100.0%**

Completed transactions represented approximately 92.0% of the dataset. Cancelled transactions accounted for around 5.0%, while returned transactions represented approximately 3.1%.
This indicates that most recorded transactions were marked as completed. Nevertheless, the presence of cancelled and returned records highlights the importance of considering order status when interpreting sales totals.
The status distribution describes the number of records in each category. It should not be interpreted as a confirmed cancellation rate for all customer orders unless the dataset's order-level structure and status definitions are verified.

9.6 Time-Based Sales Analysis

Time-based analysis examined annual and monthly variations in recorded sales to identify changes over the period covered by the dataset.

9.6.1 Annual Recorded Sales

Table 16. Recorded Sales by Year

Year	Recorded sales
2024	1,460,427
2025	1,444,771
2026 (January – June)	540,552

Recorded sales were similar in 2024 and 2025. Sales declined from 1,460,427 in 2024 to 1,444,771 in 2025, representing a decrease of approximately 1.1%.
The 2026 total was 540,552, but this figure covers only January to June. It is therefore not directly comparable with the full-year totals for 2024 and 2025. The available data does not establish whether full-year sales in 2026 will increase or decrease.

9.6.2 Monthly Sales Patterns

The monthly analysis identified May 2025 as the month with the highest recorded sales, at 130,625. February 2026 recorded the lowest monthly sales, at 75,132, among the months examined.
These results demonstrate that recorded sales varied across months rather than remaining constant throughout the observation period. The monthly comparison provides a basis for examining seasonal or recurring patterns, although a consistent seasonal trend would require further evidence across comparable periods.

The incomplete 2026 reporting period was taken into account when interpreting monthly and annual comparisons.

9.7 Quantity, Pricing and Discount Analysis

Descriptive statistics were calculated for Quantity, UnitPrice and Discount to examine their central values, variability and ranges.

Table 17. Descriptive Statistics for Quantity, Unit Price and Discount

Statistic	Quantity	UnitPrice	Discount
Count	49,204	49,204	49,204
Mean	1.88	40.29	7.48
Median	2.00	28.00	5.00
Standard deviation	1.08	42.62	7.75
Minimum	-2	7.00	0.00
25th percentile	1.00	11.00	0.00
75th percentile	2.00	58.00	10.00
Maximum	5	260.00	30.00

The mean quantity was 1.88, while the median was 2. This indicates that transaction records typically contained around two units. The negative minimum of −2 reflects the cancelled transaction retained during preprocessing and should be considered when interpreting the quantity statistics.

The mean unit price was 40.29, compared with a median of 28.00. The standard deviation of 42.62 and the range from 7 to 260 indicate considerable variation in unit prices. The difference between the mean and median suggests that higher-priced products influenced the average.

The mean discount was 7.48, while the median was 5.00. Discount values ranged from 0 to 30, with a standard deviation of 7.75. These results indicate variability in the discounts recorded across transactions.

The currency and discount units were not confirmed in the dataset documentation; consequently, no currency symbol or percentage interpretation has been assigned to these values.

9.7.1 Relationship Between Unit Price and Sales

The relationship between UnitPrice and Sales was examined using correlation analysis. The calculated correlation coefficient was approximately **0.78**, indicating a strong positive linear association.

This means that higher unit prices tended to be associated with higher recorded sales values in the dataset. However, this relationship does not establish that unit price alone caused higher sales. Quantity and other transaction characteristics may also contribute to the recorded sales value.

9.8 Relationships Between Numerical Variables

Correlation analysis was used to investigate the direction and strength of linear relationships between Age, Quantity, Discount, UnitPrice and Sales.

Table 18. Selected Correlation Results

Variable pair	Correlation coefficient	Interpretation
UnitPrice and Sales	0.78	Strong positive relationship
Quantity and Sales	0.42	Moderate positive relationship
Discount and Sales	-0.05	Very weak negative relationship
Age and Sales	Approximately 0.00	Little or no linear relationship

The strongest relationship among the selected pairs was between UnitPrice and Sales, with a correlation of 0.78. Quantity and Sales had a moderate positive correlation of 0.42, suggesting that higher quantities tended to be associated with higher recorded sales.
Discount and Sales had a correlation of −0.05, indicating almost no linear association in this dataset. Similarly, the correlation between Age and Sales was approximately zero, suggesting that age did not have a meaningful linear relationship with recorded sales.
These findings describe associations within the observed data rather than causal relationships. A weak correlation does not rule out nonlinear relationships or associations involving other variables.

9.9 Summary of EDA Findings

The EDA identified several important patterns in the e-commerce dataset. Electronics generated the highest recorded sales among product categories, while Accessories had the highest quantity sold. At the individual product level, Headphones recorded the highest sales.
Regular customers contributed the largest share of recorded sales, and Tehran had the highest sales among the cities examined. Gateway was the most frequently recorded payment method, while most transaction records were marked as completed.
Recorded sales were similar in 2024 and 2025, with monthly variations throughout the observation period. May 2025 recorded the highest monthly sales, while February 2026 recorded the lowest among the months examined. The incomplete 2026 reporting period limits direct comparisons with the previous full years.
Correlation analysis showed a strong positive relationship between unit price and sales, a moderate positive relationship between quantity and sales, and little linear association between discount or age and sales.
Overall, the EDA demonstrated that recorded sales varied across product categories, products, customer segments, cities and time periods. These results provide the analytical basis for the visualisations presented in Section 10 and the consolidated findings and recommendations discussed later in the report.

10. Data Visualisation

Following the exploratory data analysis, data visualisation was used to examine the patterns and relationships within the e-commerce dataset in a more visual and interpretable form. While the numerical analysis provided an overview of sales performance, customer characteristics and transaction activity, the visualisations helped illustrate how these patterns differed across product categories, individual products, customer segments, geographical locations and time periods.
A total of twelve visualisations were developed using Matplotlib, Seaborn and Plotly Express. These included histograms and box plots to examine data distributions, bar charts and a doughnut chart to compare sales performance and payment preferences, a line chart to explore monthly sales trends, and scatter plots and a correlation heatmap to investigate relationships between numerical variables. A grouped count plot was also used to compare order statuses across customer segments.
The visualisations are organised into univariate, bivariate and multivariate analysis according to the number and type of variables examined. Together, the charts provide a visual basis for understanding the dataset's key characteristics, comparing performance across different dimensions and identifying relationships that warrant further consideration.

10.1 Univariate Analysis

10.1.1 Distribution of Customer Ages (Histogram, Library: Seaborn)

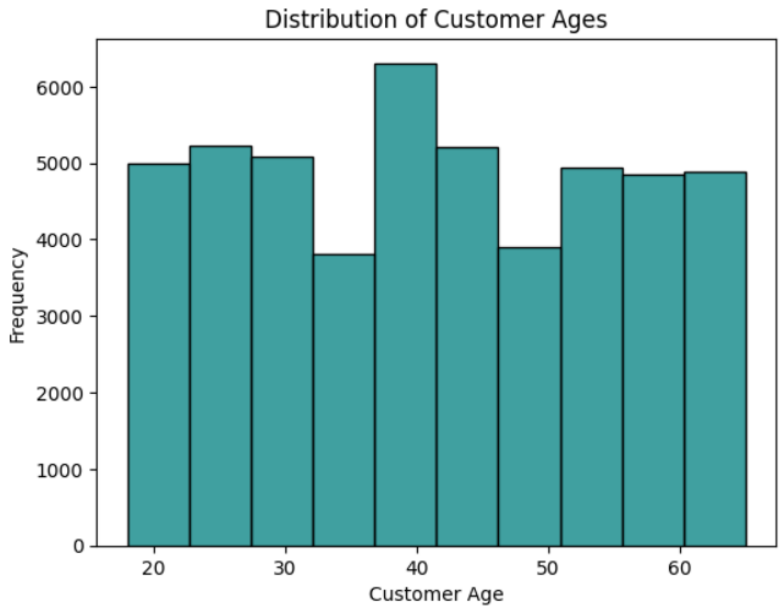


Figure 10.1: Distribution of Customer Ages

The histogram shows customer ages ranging from 18 to 65. The largest concentration of transaction records is around ages 37–42. Records are spread across the age range, with no single narrow age group dominating. As customers may appear in multiple transactions, this chart represents transaction records rather than unique customers.

10.1.2 Distribution of Sales (Box plot, Library: Seaborn)

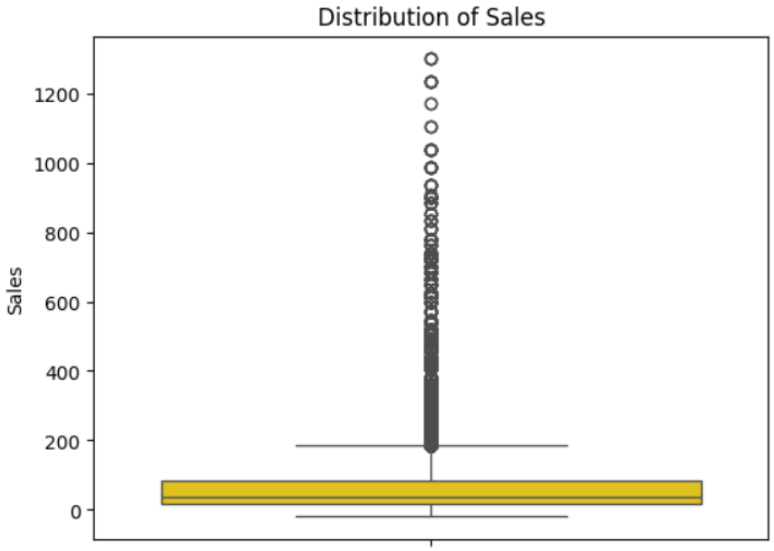


Figure 10.2: Distribution of Sales

The box plot illustrates the distribution of recorded sales values and highlights the presence of potential outliers. Most transactions fall within a comparatively lower sales-value range, while a smaller number of observations extend considerably further along the upper end of the distribution, including values above 1,200. These observations indicate variation in transaction values, with some transactions differing substantially from the typical range. However, the presence of outliers does not necessarily indicate incorrect data, and the chart alone cannot determine whether these transactions are valid or how much they contribute to total recorded sales.

10.1.3 Distribution of Payment Methods (Doughnut chart, Library: Plotly)

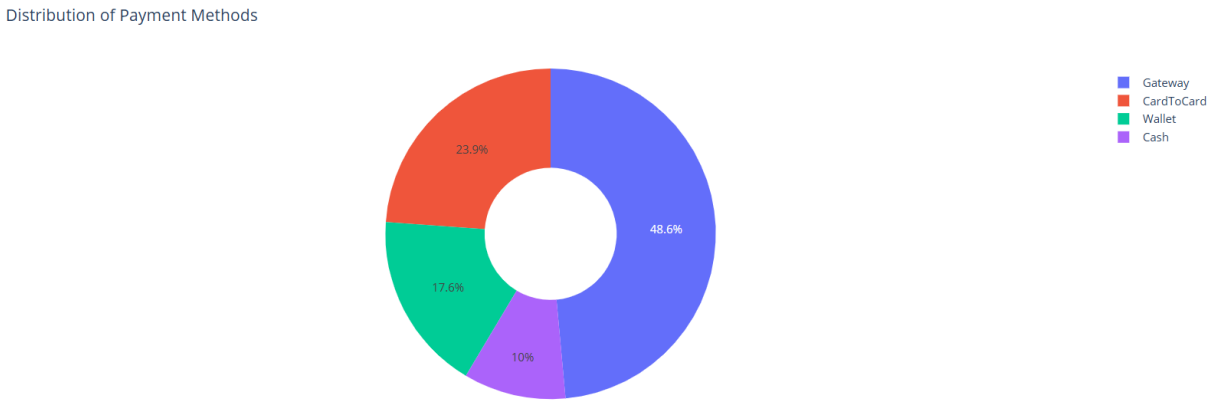


Figure 10.3: Distribution of Payment Methods

The doughnut chart shows that Gateway is the most frequently used payment method, with 23,891 transactions (48.6%). CardToCard accounts for 11,738 transactions (23.9%), followed by Wallet with 8,655 (17.6%) and Cash with 4,920 (10%). Together, Gateway, CardToCard and Wallet account for about 90% of transactions, indicating that digital payment methods are used much more often than cash in this dataset.

10.2 Bivariate Analysis

10.2.1 Total Sales by Product Category (Bar chart, Library: Matplotlib)

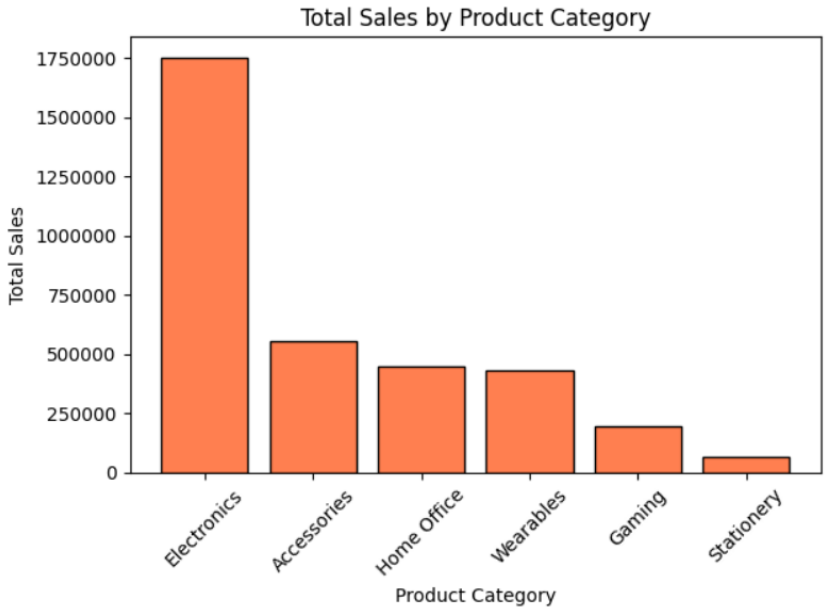


Figure 10.4: Total Sales by Product Category

Electronics generated the highest total sales, at approximately 1.75 million, accounting for around 50.8% of recorded sales. Accessories ranked second. Home Office and Wearables had similar sales levels, while Stationery recorded the lowest total sales.

10.2.2 Top 10 Products by Sales (Horizontal bar chart, Library: Plotly)

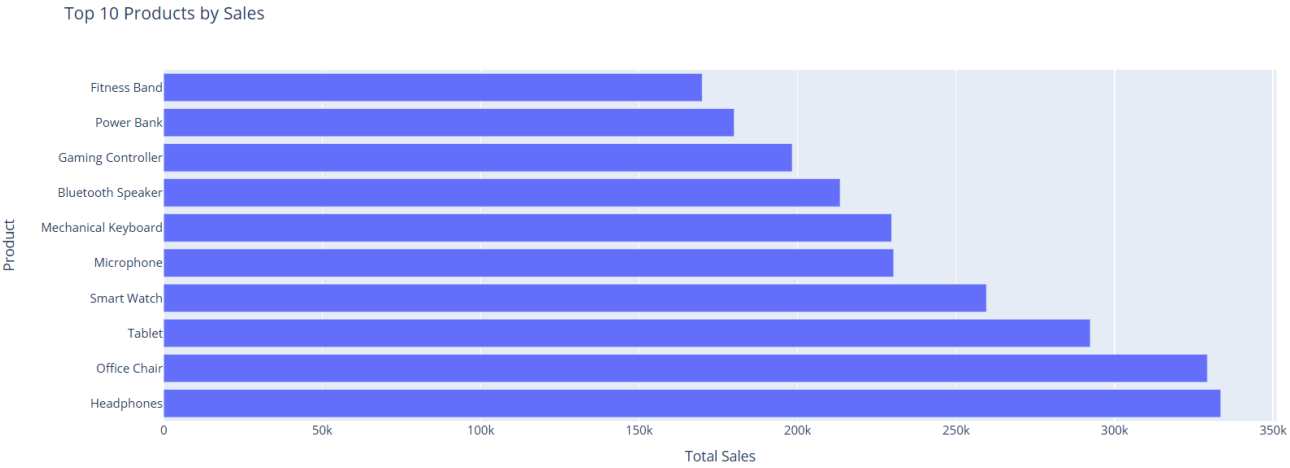


Figure 10.5: Top 10 Products by Sales

Headphones recorded the highest sales among the ten products, at 333,531. Office Chair followed at 329,292, and Tablet at 292,344. The top ten includes products from several categories, suggesting that high sales are not limited to a single product type.

10.2.3 Total Quantity Sold by Product Category (Bar chart, Library: Matplotlib)

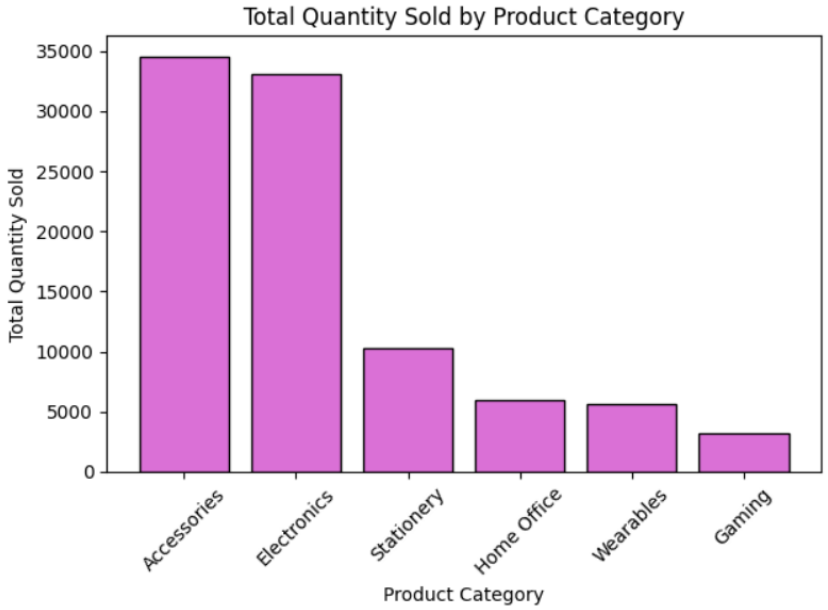


Figure 10.6: Total Quantity Sold by Product Category

Accessories had the highest quantity sold, followed by Electronics. Electronics generated the highest sales value, while Accessories sold the most units. This difference suggests that sales value is influenced by more than the number of units sold, including differences in product prices.

10.2.4 Total Sales by Customer Segment (Bar chart, Library: Matplotlib)

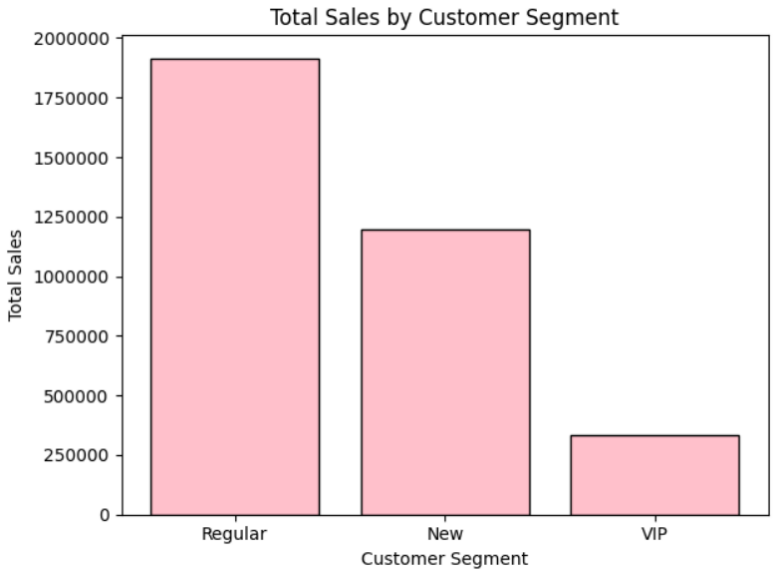


Figure 10.7: Total Sales by Customer Segment

This chart compares total recorded sales across customer segments. Regular customers account for the largest number of transactions, followed by New customers and then VIP customers. Transaction counts should not, on their own, be taken as evidence that a segment has higher average spending per customer.

10.2.5 Relationship Between Unit Price and Sales (Scatter plot, Library: Seaborn)

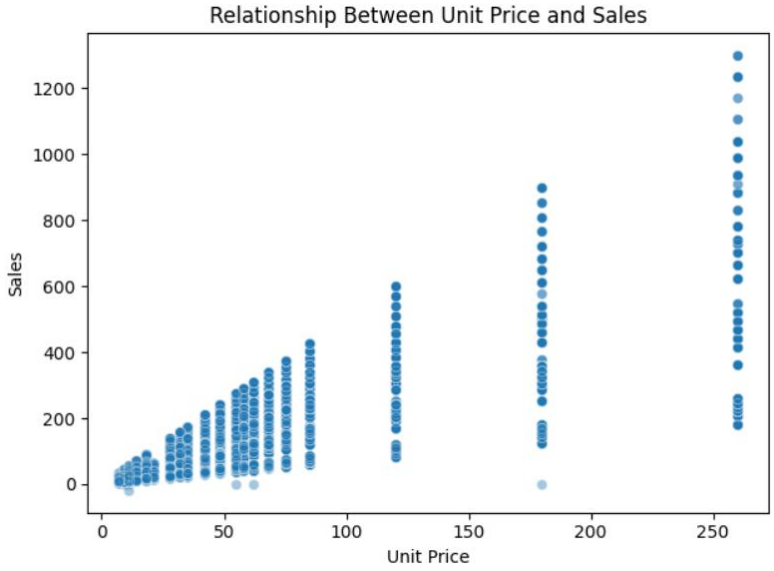


Figure 10.8: Relationship Between Unit Price and Sales

The scatter plot examines the relationship between unit price and recorded sales value across transactions. The overall pattern indicates a positive association, whereby transactions involving higher unit prices tend to be associated with higher sales values. The spread of observations also shows that transactions with similar unit prices can have different sales values, suggesting that unit price alone does not account for all variation. Differences in quantity, discounts and other transaction characteristics may also be relevant. Any observations close to or below zero should be interpreted in the context of the underlying records, particularly where unusual transaction values have been identified during data cleaning. The visual relationship indicates association rather than causation.

10.2.6 Total Sales by City (Bar chart, Library: Seaborn)

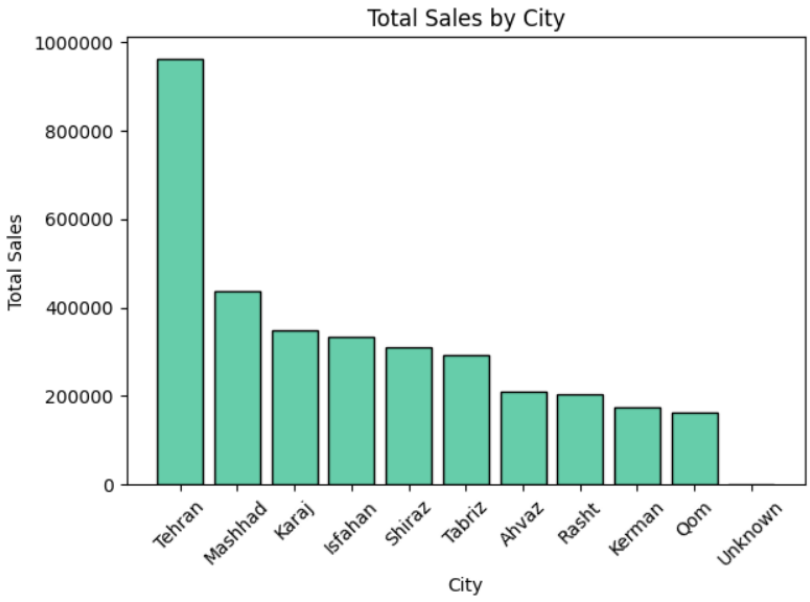


Figure 10.9: Total Sales by City

Tehran recorded the highest sales value, at 963,320, followed by Mashhad at 438,089 and Karaj at 350,030. Tehran's recorded sales value was more than twice that of Mashhad, illustrating a substantial difference between the two leading cities. Qom recorded the lowest sales value among the ten cities presented, at 162,760. These differences highlight the geographical variation in recorded sales within the dataset. However, the chart does not explain the reasons for these differences or account for variations in the number of customers represented in each city.

10.2.7 Order Status by Customer Segment (Grouped count plot, Library: Seaborn)

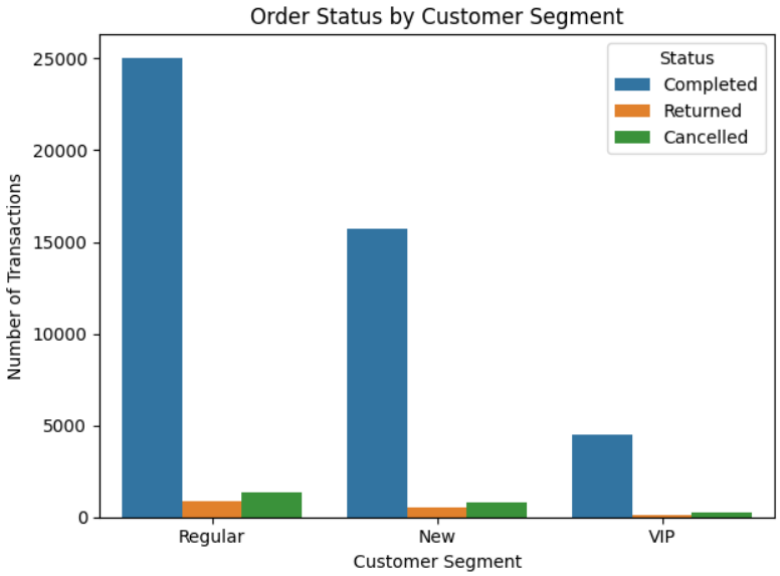


Figure 10.10: Order Status by Customer Segment

Completed transactions are the most common status across all three customer segments. Regular customers have the largest numbers of returns and cancellations, while VIP customers have the fewest. These are transaction counts, so the difference may partly reflect the greater number of Regular customer transactions overall.

10.3 Multivariate Analysis

10.3.1 Monthly Sales Trends by Year (Line chart, Library: Plotly)

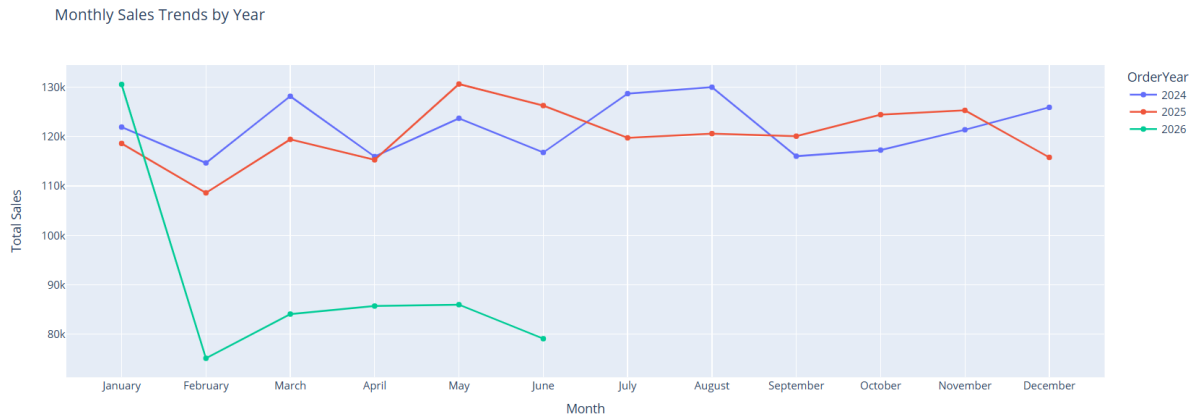


Figure 10.11: Monthly Sales Trends by Year

Monthly sales fluctuated throughout 2024 and 2025. The highest monthly value shown is May 2025, at 130,625. January 2026 recorded 130,550, the highest January value across the three years. Sales from February to June 2026 were lower than in the corresponding months of 2024 and 2025. However, the 2026 data covers only January to June, so it cannot be compared as a complete year.

10.3.2 Correlation Heatmap of Numerical Variables (Heatmap, Library: Seaborn)

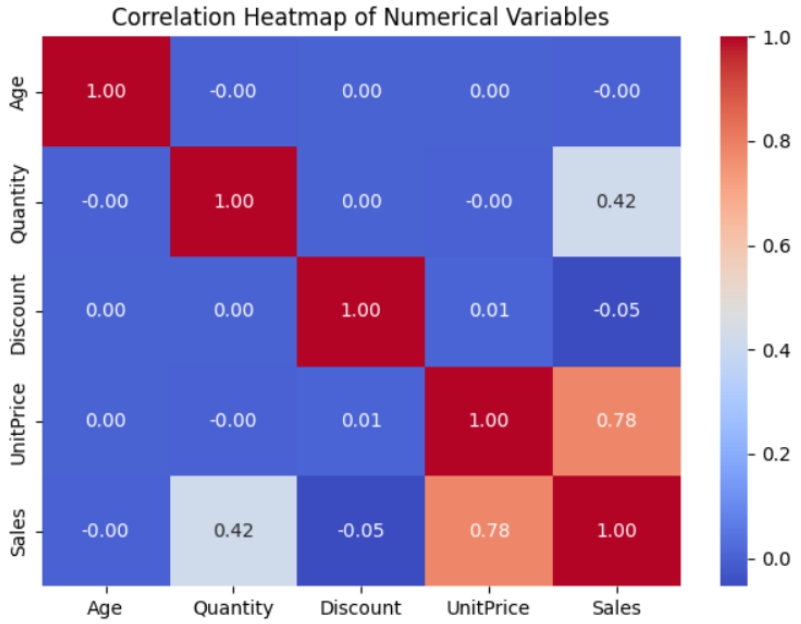


Figure 10.12: Correlation Heatmap of Numerical Variables

The heatmap shows a strong positive correlation between UnitPrice and Sales (0.78), and a moderate positive correlation between Quantity and Sales (0.42). Discount and Sales have a correlation of -0.05, indicating almost no linear relationship in this dataset. Age and Sales are also almost uncorrelated, at approximately 0.00. Correlation describes how variables move together; it does not establish that one variable causes changes in another.

11. Key Findings and Insights

- **Data quality:** After cleaning and preprocessing, the final dataset contained 49,204 rows and 20 columns, with missing values addressed and duplicate records removed. One cancelled transaction with a negative quantity remained as an unresolved anomaly.
- **Category performance:** Electronics generated the highest recorded sales value (1,751,491; 50.8%), while Accessories recorded the highest quantity sold.
- **Top-performing products:** Headphones recorded the highest sales value among the top ten products, at 333,531.
- **Customer segments:** Regular customers contributed the largest share of recorded sales (55.6%), followed by New customers (34.7%) and VIP customers (9.7%).

- **Geographical performance:** Tehran recorded the highest sales value (963,320), more than twice that of Mashhad (438,089).
- **Payment preferences:** Gateway was the most frequently recorded payment method, accounting for 48.6% of transactions.
- **Order status:** Completed transactions represented approximately 92% of records, with cancellations and returns accounting for much smaller shares.
- **Sales trends:** Annual recorded sales were broadly similar in 2024 and 2025. Monthly sales fluctuated, with May 2025 recording the highest monthly value (130,625). The 2026 figures cover only January to June.
- **Numerical relationships:** UnitPrice had the strongest positive correlation with Sales (0.78), followed by Quantity (0.42). Age and Discount showed little to no linear relationship with Sales.

12. Recommendations

- Optimise product category performance: Monitor Electronics sales closely, given its substantial contribution to recorded sales. Review the performance of lower-contributing categories, such as Stationery, to identify opportunities for improvement.
- Improve inventory planning: Consider both sales value and quantity sold when planning stock levels. The high quantities recorded for Accessories suggest that demand should be considered alongside unit prices to avoid stock shortages or excess inventory.
- Strengthen customer segment analysis: Investigate purchasing frequency, average transaction value and repeat-purchase behaviour across Regular, New and VIP customers. This would help explain differences in segment contributions and identify opportunities to improve customer retention.
- Investigate geographical sales patterns: Examine customer numbers, purchasing behaviour and product preferences in Tehran and other cities to understand the differences in recorded sales and identify potential areas for growth.
- Monitor payment methods: Ensure that frequently used payment methods, particularly Gateway, remain reliable and convenient. Review transaction success and failure rates to identify potential payment-related issues.
- Review cancellations and returns: Examine the reasons for cancelled and returned transactions to identify recurring issues and opportunities to improve the purchasing and fulfilment process.
- Investigate monthly sales fluctuations: Compare sales across equivalent months and examine possible influences, such as promotions, product availability and customer demand, before drawing conclusions about seasonal patterns.
- Conduct further sales analysis: Investigate the relationship between unit price, quantity and sales value in greater detail. Incorporating product costs, profit margins and discount information would help assess profitability rather than relying on sales value alone.
- Maintain data quality: Continue validating transaction records, removing confirmed duplicates, standardising categorical values and investigating unusual quantities or sales values. Particular attention should be given to unresolved anomalies, such as cancelled transactions with negative quantities.

13. Conclusion

This project aimed to analyse e-commerce sales data to understand customer behaviour, product performance and sales trends. Using Python in Google Colab, the project followed a structured approach involving data cleaning, preprocessing, exploratory data analysis and data visualisation to examine the dataset and identify meaningful patterns.

The data preparation process involved addressing missing values, removing duplicate records, standardising categorical information, converting date columns into appropriate formats and investigating unusual transaction values. Additional features were created to support analysis across different time periods and customer age groups. Following preprocessing, the final dataset contained 49,204 rows and 20 columns, providing a structured foundation for further analysis.

The exploratory data analysis and visualisations revealed several important patterns. Electronics generated the highest recorded sales value, while Accessories accounted for the greatest quantity sold. Regular customers contributed the largest share of recorded sales, and Tehran recorded the highest sales value among the cities analysed. The analysis also identified differences in payment-method usage, variations in monthly sales and positive relationships between UnitPrice, Quantity and Sales, particularly the strong positive correlation between UnitPrice and Sales.

The project achieved its main objectives by preparing the dataset for analysis, examining sales and customer characteristics, identifying patterns through visualisations and developing practical recommendations based on the findings. It also provided an opportunity to apply data analysis techniques using Python libraries, including Pandas, Matplotlib, Seaborn and Plotly, strengthening practical skills in data preparation, exploratory analysis and visual communication.

Overall, the project demonstrated how data analysis and visualisation can transform raw transaction records into meaningful insights about product performance, customer segments and sales patterns. The findings provide a foundation for understanding the information contained in the dataset and show how analytical techniques can support evidence-based decision-making in e-commerce.

14. Limitations and Future Scope

14.1 Limitations

Although the project identified useful patterns in e-commerce sales data, several limitations should be considered when interpreting the findings.

- Limited dataset information: The analysis was based on the variables available in the dataset. Information such as product costs, profit margins, marketing expenditure and customer feedback was not included, limiting the assessment of overall business performance.
- Incomplete data for 2026: The dataset contained records for January to June 2026 only. Therefore, comparisons between 2026 and the complete years of 2024 and 2025 should be interpreted cautiously.
- Limited customer behaviour analysis: The dataset supported comparisons between customer segments, but further information on individual purchase histories and customer retention would be needed to understand long-term purchasing behaviour.
- Correlation does not establish causation: Relationships between variables such as UnitPrice, Quantity and Sales indicate associations within the dataset but do not establish that one variable directly causes changes in another.

- Remaining data anomaly: One cancelled transaction with a negative quantity remained in the cleaned dataset. Although the record was retained because its validity could not be conclusively established, it may affect quantity-related statistics.

14.2 Future Scope

The project could be extended in several ways to provide a more detailed understanding of e-commerce performance.

- Analyse profitability: Incorporating product costs and profit margins would help identify which products and categories generate the greatest profit, rather than relying only on sales value.
- Study customer purchasing behaviour: Analysing repeat purchases, purchase frequency and customer retention over time could provide a clearer understanding of customer loyalty and differences between customer segments.
- Investigate cancellations and returns: Examining the reasons for cancelled and returned orders could help identify recurring problems and opportunities to improve the purchasing and fulfilment process.
- Extend sales trend analysis: Including additional months and future years would support more complete year-on-year comparisons and help investigate possible seasonal patterns.
- Improve geographical analysis: Combining sales records with customer numbers and additional geographical information could help explain differences in sales across cities.
- Develop predictive analysis: With sufficient historical data, future work could explore sales forecasting and demand prediction to support inventory planning and business decisions.

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